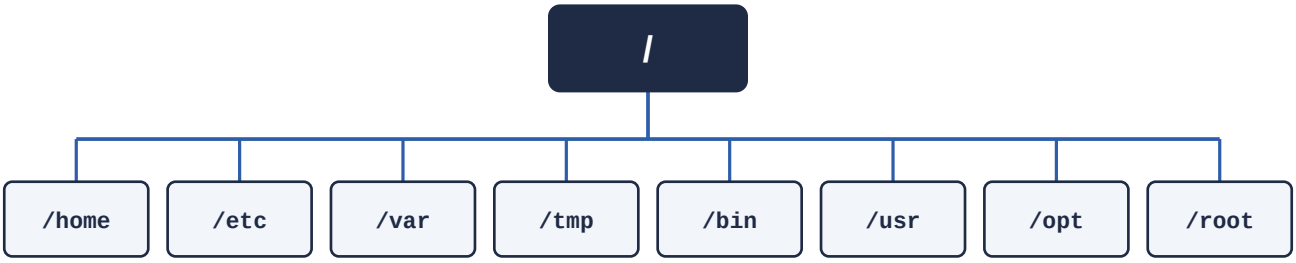
Linux File System

Core directories, the folder hierarchy, and the commands used to explore them — Task 4

Folder Hierarchy



*All eight directories sit directly beneath **/**, the root of the entire file system. Unlike Windows, Linux has no separate drive letters (C:\, D:\) — every file and folder, regardless of which physical disk it lives on, is reached by branching down from this one root.*

Core Directories Explained

Folder	What It's For
`/`	Top level of the entire file system. Every other file and folder is located somewhere beneath it — there is no separate drive letter like Windows' C:\; everything branches from here.
`/home`	Personal directory for each regular user (e.g. /home/alice), holding that user's documents, downloads, and personal settings.
`/etc`	System-wide configuration files — e.g. passwd, hosts, fstab — that control how the OS and installed programs behave.
`/var`	"Variable" data that changes while the system runs, such as log files (/var/log), caches, and spool queues.
`/tmp`	Temporary files created by running programs. Contents are short-lived and are often cleared automatically on reboot.
`/bin`	Essential command binaries needed for basic system operation, available to every user — e.g. ls, cp, mv, bash.
`/usr`	The bulk of installed programs, libraries, and documentation ("Unix System Resources"). Despite the name, it is not the same as /home.
`/opt`	Optional, third-party, or add-on software packages that are not part of the base operating system install.
`/root`	The administrator (root) user's personal home directory, kept separate from /home for security and system-recovery reasons.

Commands Used to Explore Each Folder

Command	What It Does
`cd`	Change Directory — moves into the folder you specify, making it your new current location.
`pwd`	Print Working Directory — confirms the full path of where you currently are; useful right after cd to verify you landed in the right place.
`ls`	List — shows the files and subfolders inside your current location once you've arrived.

The same three-command pattern is repeated for every folder in the hierarchy above:

```
cd /etc
pwd
ls
```

*Move in with **cd**, confirm the new location with **pwd**, then list what's inside with **ls** — then repeat for the next folder on the list.*

Prepared for DEVOPS-001 — Task 4: Linux File System.